

Trading, the loss of transactional benefits, and pricing of a cryptocurrency

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Abstract

This study explores a critical and underexamined issue in cryptocurrency economics: the forfeiture of transactional benefits—such as lower costs, global accessibility, and anonymity—when cryptocurrencies are used for trading rather than for transactions. We question why rational investors engage in cryptocurrency trading despite this loss, and how such behavior affects price dynamics. Departing from traditional literature that attributes cryptocurrency valuation to fundamentals and sentiment, we develop a single-agent model—drawing from Biais et al. (2023)—to explicitly incorporate the loss of transactional benefits. Our model reveals that for trading to occur, investors must hold a minimum level of optimism regarding future prices, making this sentiment a prerequisite rather than a consequence of market behavior. We demonstrate that increased trading, relative to transactional use, elevates cryptocurrency prices only if sentiment surpasses this critical threshold. Conversely, under constant price equilibria, heightened investor allocation to cryptocurrency trading leads to declining crypto prices and rising standard currency prices. This work offers a novel theoretical framework for understanding the role of optimism and lost utility in cryptocurrency markets and paves the way for empirical validation and multi-agent extensions.

1. Introduction.

Using cryptocurrency to buy goods and services or other financial assets, such as ICO, brings transactional benefits, including lower transaction costs, global accessibility, and anonymity. These transactional benefits are lost when a cryptocurrency is bought to sell in future for trading purposes, which raises very important questions. Why will an investor buy a cryptocurrency for trading purposes, given the loss of the transactional benefits? How will the price of a cryptocurrency be adjusted with the loss of the transactional benefits to the traders?

Although there is a lot of literature that shows the pricing of a cryptocurrency is driven by fundamentals, i.e., transactional benefits and investors' sentiments, our study addresses the roles of transactional benefits and sentiments in pricing from a completely different perspective. Using the model of Biais et (2023), we show that loss of transactional benefits in trading must be compensated with a minimum optimism level about price expectations as a participatory condition for the trading. Hence, we show that trading in cryptocurrency comes along with an optimistic sentiment in the market, which is not a necessary condition in the existing literature.

Unlike other literature where the users and noise traders are treated as different agents, we deliberately choose a single-agent investor model to model the loss of transactional benefits explicitly. We also show that the loss of transactional benefits will be compensated with an optimist take on prices in each period, leading to the formation of an asset price bubble in the cryptocurrency. We also show that an increase in the trading of the cryptocurrency over using it for transactions will increase its price if only the level of investor sentiments is above the minimum optimism; otherwise, it has no impact.

Under the constant price equilibria, we show that the cryptocurrency prices fall and prices of a standard currency increase as the investors' allocation to cryptocurrency for trading purposes increases.

2. Literature Review:

Initially, a cryptocurrency was viewed as an alternative currency; Nakamoto (2008) defines a bitcoin as a peer-to-peer electronic cash system facilitating direct online payments between parties without involving financial institutions. However, it also functions as a financial asset. While a currency has properties of the medium of exchange, a unit of account, and value storage, financial assets lack the first two traits, setting them apart.

The appeal of using virtual currencies (like Bitcoin) as a medium of exchange lies in low transaction costs, global accessibility, and anonymity, especially for purchasing specific goods, illegal ones. Nevertheless, factors like low confidence, acceptability, or high price volatility can deter potential users. When seen as an investment, demand for Bitcoin influences its price volatility, impacting user adoption.

When it comes to whether a cryptocurrency is used as a currency (primarily as a medium of exchange) or investment asset, the question is rather of how much. Baur et al. (2018) show that the predominant use of Bitcoins is for speculative investment rather than as a viable alternative currency or medium of exchange by examining transaction data.

Yet, classifying a cryptocurrency into a specific asset category poses a challenge. For example, Bouri et al. (2017a, 2017b) and Corbet et al. (2018c) indicate bitcoin's weak correlation with both high-risk financial assets and safe-haven assets, implying its position in a distinct and uncorrelated asset class. Additionally, Kristoufek (2013) argue that Bitcoin cannot be priced through the conventional financial model, i.e. the future cash flow model.

Nonetheless, the emerging literature agrees on some stylized facts of a cryptocurrency. The first and foremost empirical stylized fact is that the cryptocurrency market returns can only be explained by crypto market-specific factors, not by any other market (i.e., stock, bonds, commodities, currency) and economic factors. (Liu et al., 2018; Liu & Tsyvinski, 2021). There is also no role of supply-side factors, such as the cost of mining, in explaining the return (Liu & Tsyvinski, 2021); this has emerged as a second stylized fact. The third fact is the presence of high volatility and non-normality of the returns of cryptocurrencies (Kim et al., 2021). The fourth fact is the presence of momentum in the prices of cryptocurrency, which is also widely supported by the literature. (Liu et al., 2018; Liu & Tsyvinski, 2021).

In a recent paper, Liu et al. (2022) conducted a comprehensive study of factors that could explain the crypto market returns and identified three factors- crypto market, size, and momentum- that fully explain the return. They show that the momentum effect is driven by speculation.

A cryptocurrency is also susceptible to price bubbles, driven by investors seeking speculative profits through currency exchange. Investor speculation, often disconnected from a currency's intrinsic value, can significantly impact the price that non-investors (users) must pay to use the currency for transactions. Unlike traditional currencies, where governments can adjust the supply to counter bubbles, cryptocurrencies usually follow a predetermined supply trajectory, making them more vulnerable to price bubbles(see, for example, Yermack 2015; Kjaerland et al.,2018; Patel, Tanwar, Gupta, & Kumar, 2020).

To sum up, Cryptocurrency serves both currency and financial assets. It offers low-cost transactions but faces volatility and acceptance issues. Primarily used for investment, its classification remains challenging.Prices of cryptocurrency is mainly driven by speculative behaviour, exposing the cryptocurrency to asst price bubbles.

3. Theoretical Framework & Model

Our theoretical framework is an extension of the basic model of Biais et (2023), which is itself built upon the work of Garratt and Wallace (2018). They employ Wallace's (1981) multi-currency model to investigate how the prices of a cryptocurrency and a traditional currency are jointly determined. Their analysis considers the occurrence of a potential market crash triggered by a sunspot, which Biais et (2023) exploited to drive constant price equilibria. We also limit our analysis to the constant price equilibria given the transactional benefit loss when the investors are using some of the currency not for the transactional purpose.

We built on the model of Biais et (2023). We introduce noise trading behaviour in cryptocurrency investors. We explicitly model the speculative behaviour of the investors based on their noisy expectations about future cryptocurrency prices. In our model, the investors buy the cryptocurrency for both transactional benefits and speculative purposes. Whereas in the Biais et al. (2023) model, they only buy for transactional purposes.

We use the overlapping generation model, in which infinite time is divided into discrete periods. The overlapping generation model is widely used in micro-found money (see for example, Samuelson, 1958, Wallace,1980, Tirole, 1985). Recently, this has also been used in cryptocurrency(Biais et,2023; Saleh (2020); Garratt and Wallace, 2018).

There is one consumption good and three assets- a cryptocurrency, a standard currency, and a risk-free asset. There are three types of agents in the market: investors, miners and hackers. All agents are price takers. At the beginning of the model, there is a generation of old investors, miners and hackers, holding the supply of cryptocurrency X_1 and standard currency m .

A representative investor is born as young with the endowment of e_t units of consumption goods in period one, with the following budget constraint, eq (1). From the endowment, he buys q_t units of cryptocurrency at the price of p_t , $\hat{q}_t$ units of standard currency at the price of $\hat{p}_t$, and saves s_t . Whatever left is his consumption in period one, c_t^u . φ_t is the cost of buying cryptocurrency, and it is linearly linked with the value of cryptocurrency.

$$c_t^y = e_t - s_t - q_t p_t (1 + \varphi_t) - \hat{q}_t \hat{p}_t \quad (1)$$

He buys the coins in the proportion of $(1 - \tau_t)$ for transactional purpose and others in proportion τ_t for speculative purposes. We assume that the investor perceives these two allocations for the coins as separate accounts, drawing from the literature on the mental accounting. He expects to get φ_{t+1} transactional benefit on the quantity that he buys for this purpose. We assume that these transactional benefits stem from the transactions of the common goods among the network of cryptocurrency users, as shown by Sockin and Xiong (2023). We assume $\varphi_{t+1} \geq -1$ to avoid degenerate case(Biais et ,2023).

As an incentive to speculate, he expects the future price to be higher by ρ_{t+1} , which represents over-optimism on his part. h_{t+1} is the proportion of the coins expected to be stolen by the hackers during the period t to $t+1$. Hence, his budget constraint when he becomes old at $t+1$ is shown by eq (2).

$$c_{t+1}^o = s_t(1 + r_t) + (1 - h_{t+1})(1 + \rho_{t+1})\{(1 - \tau_t)q_t p_t(1 + \theta_{t+1}) + \tau_t q_t p_t\} + \hat{q}_t \hat{p}_t \quad (2)$$

The budget constraints of miners and hackers are shown in eq (3) and eq (4) respectively. The introduction of miners accounts for the creation of new coins (supply side), whereas the hackers are included to consider the possibility of hacks. However, their actions are kept minimal because our main focus is on the investors' behaviour. Thus, new miners and hackers are born, perform their activities and consume after selling their cryptocurrencies.

$$c_{t+1}^m = (X_{t+1} - X_t)p_{t+1} + \varphi_{t+1}q_{t+1}p_{t+1} \quad (3)$$

$$c_{t+1}^h = h_{t+1}q_{t+1}p_{t+1} \quad (4)$$

We assume that utilities in two different periods are additive and separable, with $u' > 0$ and $u'' \leq 0$. $\{X_t\}_{t>0}$, $\{\varphi_t\}_{t>0}$, $\{\theta_t\}_{t>0}$, & $\{\pi_t\}_{t>0}$, are assumed to be exogenous processes, following Biais et (2023). We also assume $\{\tau_t\}_{t>0}$ to be exogenous process, that means how much the investors buy the cryptocurrency for transactional purpose and how much for trading purpose in each period depends upon exogenous factors, such as government regulations.

π_t is the probability of crash, when the cryptocurrency price permanently falls to zero. It is assumed to be purely extrinsic random variable, caused by an extrinsic change in beliefs due to a sunspot, as in the model of Garratt and Wallace (2018).

4. Pricing Cryptocurrency under General Preferences

Here we are driving the equilibrium price of cryptocurrency in general equilibrium setting, given the noise trading behaviour of the investors.

In the equilibrium setting, a young investor in period t solves:

$$\max_{q_t, s_t, \hat{p}_t} \{u(c_t^y) + \beta E_t u(c_{t+1}^o)\} \quad (5)$$

$$\text{Subject to } c_t^y \geq 0, (1) \text{ and } (2) \quad (6)$$

In each period t , the market clears at: $q_t = X_t, \hat{q}_t = m$, and $s_t = 0$

The first order optimality condition, 1st partial derivative of eq (5), with respect to q_t results in to:

$$p_t = \beta E_t \left[\frac{\dot{u}(c_{t+1}^o)}{\dot{u}(c_t^y)} (1 + \rho_{t+1})(1 - h_{t+1}) \frac{\{1 + \theta_{t+1}(1 - \tau_t)\}}{(1 + \varphi_{t+1})} p_{t+1} \right] \quad (7)$$

Eq (7) also holds when the condition that $c_t^y \geq 0$ also binds, in other words, if $c_t^y = 0$.

When crash risk is considered, eq (7) results into:

$$p_t = \beta(1 - \pi_t) E_t \left\langle \frac{\dot{u}(c_{t+1}^o)}{\dot{u}(c_t^y)} (1 - h_{t+1})(1 + \rho_{t+1}) \frac{\{1 + \theta_{t+1}(1 - \tau_t)\}}{(1 + \varphi_{t+1})} p_{t+1} \middle| \text{no crash} \right\rangle \quad (8)$$

it is important to note that price at time period t in eq (8) above is conditional on no crash environment, whereas it is not in eq (7).

The first order optimality condition with respect to s_t turns out to be:

$$\beta = \frac{1}{1+r_t} \frac{\dot{u}(c_t^y)}{E[\dot{u}(c_{t+1}^o)]} \quad (9)$$

Substituting β from eq (9) into eq (8) yield our 1st proposition:

PROPOSITION 1: *In period t, the equilibrium price of the cryptocurrency is:*

$$p_t = \frac{1}{1+r_t} E_t \left[\frac{\dot{u}(c_{t+1}^o)}{E[\dot{u}(c_{t+1}^o)]} (1 - h_{t+1})(1 + \rho_{t+1})(p_{t+1} + \theta_{t+1}(1 - \tau_t)p_{t+1}) \right] \quad (10)$$

or for an arbitrary $K > 1$,

$$p_t = E_t \left\{ \left[\left(\prod_{j=1}^K \frac{(1+\rho_{t+j})(1-h_{t+j})}{(1+r_{t+j-1})(1+\varphi_{t+j-1})} \frac{\dot{u}(c_{t+j}^o)}{E[\dot{u}(c_{t+j}^o)]} \right) p_{t+K} \right] + \left\{ \sum_{h=1}^K \left(\prod_{j=1}^h \frac{(1+\rho_{t+j})(1-h_{t+j})}{(1+r_{t+j-1})(1+\varphi_{t+j-1})} \frac{\dot{u}(c_{t+j}^o)}{E[\dot{u}(c_{t+j}^o)]} \right) \theta_{t+h}(1 - \tau_{t+h-1})p_{t+h} \right\} \right\} \quad (11)$$

When in equation (10), on period ahead future value of the cryptocurrency is replaced with formula, and this process is done iteratively, the equation (10) turns into equation(11).

Eq (10) says the price of cryptocurrency in period t is the present value of the expectation of product of four terms. The first component represents the pricing kernel, indicating the relationship between the marginal utility of consumption and the cryptocurrency price. The second element accounts for the vulnerability to hacks, whereas third term counts for the optimism. Lastly, the fourth term covers the total of the cryptocurrency price in period t + 1 and its adjusted net transactional benefit, adjusted with transactional benefit loss.

When an investor buys some portion of cryptocurrency for trading purpose, it affects the price in two ways in each period. Firstly, he loses the transactional benefit in proportion to his investment for trading purpose, $\theta_{t+1}(\tau_t)$. As a result, he receives the adjusted transactional benefits, equals to $\theta_{t+1}(1 - \tau_t)$, as it is shown in eq (10). Secondly, as incentive to participate in trading, he takes the optimism take the expected price of cryptocurrency by the factor of ρ_{t+1} . In the next section, we will discuss in detail on these.

5. Loss of Transactional Benefits and Optimism

As it can be seen in eq (10), when some fraction of cryptocurrency (τ_t) is used for trading by an investor, the investors lose the transactional benefit on that proportion, resulting in to adjusted transactional benefit of $\theta_{t+1}(1 - \tau_t)$. In other words, this adjusted transactional benefit is the benefit on the proportion of investment for the transactional purpose.

It is very important to realize that an investor expects to get different benefits from the same currency depending upon how it is used. He receives transactional benefits only when using the cryptocurrency for transactions. If used for trading, he won't receive any transactional benefit; instead, he'll lose it. This raises the very question: why would the investor buy the cryptocurrency for the trading purpose, given he will lose transactional benefit?

Before answering this question, we need to consider two facts. Firstly, the transactional benefits of cryptocurrency are very limited, questioning the viability of cryptocurrency as a widely accepted medium of exchange. One of the reasons for the limited usage is due to its partial adoption limited common goods that can be traded with it, see for instance Sockin and Xiong (2023). Another one major reason is the government restrictions. Secondly, the cryptocurrency is more than often used for speculative perspective rather than as the medium exchange (Baur et al., 2018).

Given these two facts, it becomes quite sensible to assume that investors buy the cryptocurrency more than what is required for transactional purpose. However, to compensate for the loss of transactional benefit on the proportion of trading purpose, the investor would over-expect the future price of the cryptocurrency by the $(1 + \rho_{t+1})$, as optimism adjustment, resulting in to proposition 2.

PROPOSITION 2:

In each period, the minimum optimist adjustment as incentive to the investors for the loss transactional benefit when he cryptocurrency for trading purpose in τ_t proportion is:

$$1 + \bar{\rho}_t = \frac{(1+\theta_t)}{\{1+\theta_t(1-\tau_t)\}} \quad (12)$$

Where $\bar{\rho}_t > 0 \iff \theta_t > 0$ & $0 \leq \tau_t \leq 1$, in the case when transaction benefits are positive and no short selling of the cryptocurrency and borrowing is allowed.

Proposition 2 can be also interpreted as the participation constraint for buying the cryptocurrency for trading purpose. Moreover, since the loss of transactional benefits will be compensated with an optimist take on prices in each period, an asset price bubble will form and it will explode at the rate of $\bar{\rho}_t$ per period.

PROPOSITION 3:

The change in the proportion of τ_t has no impact on the price of cryptocurrency if $\rho_t = \bar{\rho}_t$.

However, In the case of $\rho_t > \bar{\rho}_t$, higher the τ_t , higher the price.

The proposition 3 states the impact of how much cryptocurrency is used for trading purpose on its price depends upon the prevailing sentiment in the market. If the overall investor sentiment is above a minimum optimistic threshold, an increase in trading proportion drives up the cryptocurrency's price.

However, if the investor sentiments is at the minimum optimistic level, the increased trading proportion will not impact its price. Instead, in the period, the price remains the same irrespective of how much the trading proportion changes.

This observation underscores the critical role of investor sentiments in dictating the relationship between trading volumes and the price of a cryptocurrency. It suggests that investor perceptions, emotions, and prevailing market sentiments play a substantial role in determining the price movements based on the nature of cryptocurrency usage—whether primarily for trading or actual transactions

6. Constant Price Equilibrium with Risk Averse Agents

Following Biais et (2023), we drive the constant price equilibria. In which, the prices of cryptocurrency and a standard currency are constant until crash happens, The probability of a crash is also constant π . For simplicity, we also assume that the utility function of an investor is the logarithmic, this assumption won't affect the relationships that we want to draw. Moreover, the results could also valid for volatile price equilibria, the case in which the prices of prices of cryptocurrency and a standard currency are not constant, and the probability of crash is stochastic.

To simplifies our analysis, we also assumee that endowments, cryptocurrency supply, proportion invested for trading purpose, costs, and benefits remain constant. The constant price of a cryptocurrency and a standard currency is shown by p and $\hat{p}$ respectively; whereas $\hat{p}_c$ represents the price of a standard currency after the crash happens.

Under the constant price conditions, the solution to eq(5) subject to eq(6) follows.

1st order condition with respect to cryptocurrency,

$$\dot{u}(e - X(1 + \varphi) - m\hat{p}) = \beta(1 - \pi)(1 - h)(1 + \rho) \frac{(1 + \theta(1 - \tau))}{(1 + \varphi)} \dot{u}((1 - h)(1 + \rho)Xp(1 + \theta(1 - \tau)) + m\hat{p}) \quad (13)$$

1st order condition with respect to standard currency

$$\dot{u}(e - X(1 + \varphi) - m\hat{p})\hat{p} = \beta \left[(1 - \pi)\dot{u}((1 - h)(1 + \rho)Xp(1 + \theta(1 - \tau)) + m\hat{p})\hat{p} + \pi\dot{u}(m\hat{p})\hat{p}_c \right] \quad (14)$$

Substituting the right side of (13) in the left side of (14) results into

$$\dot{u}((1 - h)(1 + \rho)Xp(1 + \theta(1 - \tau)) + m\hat{p}) = \frac{\pi}{(1 - \pi)} \frac{\hat{p}_c}{\hat{p}} \frac{(1 + \varphi)}{(1 - h)(1 + \rho)((1 + \theta(1 - \tau)) - (1 + \varphi))} \dot{u}(m\hat{p}) \quad (15)$$

Rearranging eq(13) results into

$$\dot{u}((1-h)(1+\rho)Xp(1+\theta(1-\tau))+m\hat{p}) = \frac{(1+\varphi)}{\beta(1-\pi)(1+\rho)(1+\theta(1-\tau))} \dot{u}(e - X(1+\varphi) - m\hat{p}) \quad (16)$$

After the crash, $\hat{p}_c$ is defined by following equilibrium condition

$$\dot{u}(e - m\hat{p}_c) = \beta \dot{u}(m\hat{p}_c) \quad (17)$$

Solution to equations 15 and 16 under log utility comes with following proposition.

PROPOSITION 4:

Under the conditions if the investors have log utility and

$$(1 + \theta(1 - \tau)(1 - \pi) > 1 \quad (18),$$

there is an equilibrium where capitalization of cryptocurrency as a percentage of total endowment is

$$\frac{Xp}{e} = \frac{\beta}{1+\beta} \left(1 - \pi \left(1 + \frac{1}{\theta(1-\tau)} \right) \right) \quad (19),$$

whereas the capitalization of a standard currency is

$$\frac{m\hat{p}}{e} = \frac{\beta}{1+\beta} \left(\frac{\pi(1+\theta(1-\tau))}{\theta(1-\tau)} \right) \quad (20)$$

Proposition 4 demonstrates that as crash risk (π) increases, the cryptocurrency price declines, whereas the standard currency price rises. On the other hand, with increasing adjusted transactional benefits $\theta(1 - \tau)$, cryptocurrency price increases, whereas the standard currency price falls.

What's particularly intriguing to our analysis is that when investors allocate more to cryptocurrency for trading, the cryptocurrency prices drop while the prices of a standard currency rise.

More interestingly, as the investors allocation to cryptocurrency for the trading purpose increases, the cryptocurrency prices falls and prices of a standard currency increase.

7. Conclusion

Cryptocurrency offers transactional benefits when used as currency to buy goods and assets. However, when used for trading purposes, these benefits are forfeited, prompting questions about the rationale behind the trading in it. While existing studies attribute cryptocurrency pricing to fundamentals and sentiments, our research takes a unique perspective, highlighting the need for a minimum optimistic take on the future price of a cryptocurrency due to the loss of transactional benefits. This optimism, contrary to prior literature, becomes an integral part of cryptocurrency trading.

In our study, we model the explicit loss of transactional benefits using a single-agent investor approach, revealing that this loss is compensated by an optimistic outlook on prices, leading to potential asset price bubbles. We demonstrate that increased cryptocurrency trading versus transactional use affects its price positively only when investor sentiment surpasses a minimum threshold. Under constant price equilibria, our findings reveal that increased investor allocation to cryptocurrency trading leads to declining cryptocurrency prices and increasing standard currency prices.

This study is an initial theoretical attempt to model the loss of transactional benefits. It can be extended to the multi-agent model, and its propositions can also be empirically tested in future studies.

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